THE REVERSE DEGREE-POLYNOMIAL RANGE CLAUSE FOR THE $2 \times n$ MIURA-ORI FLIP GRAPH FAILS FOR EVERY $a \geq 7$

Abstract. Let v_n^d be the number of vertices of degree d in the origami flip graph $\mathrm{OFG}(M_{2,n})$ of the $2 \times n$ Miura-ori, and put $f_a(n) := v_n^{2n-a}$. A labelled theorem of Christensen, Hull, O'Neil, Pappano, Ter-Saakov and Yang, printed with no proof, asserts that f_a agrees with a polynomial of degree $\lfloor a/2 \rfloor$ on the whole of $n \geq \lceil a/2 \rceil + 1$, for every $a \in \mathbb{N}_0$. We refute the range clause. The witness at $a = 7$ consists of five integers printed in that paper's own degree-distribution table, $f_7(5), \ldots, f_7(9) = 12, 88, 296, 680, 1288$, whose fourth forward difference is $4 \neq 0$ although the window lies inside the asserted range. A generating-function argument then shows the failure is an infinite family: writing $k_a = \lfloor a/2 \rfloor$ and $G_a = (1-x)^{k_a+1} \sum_{n \geq 1} f_a(n) x^n$, we prove G_a is a polynomial of degree $3\lfloor a/2 \rfloor - 1$ or $3\lfloor a/2 \rfloor$ according to the parity of a, with leading coefficient exactly $4(-1)^{\lfloor a/2 \rfloor + 1}$ and $G_a(1) > 0$. Hence the asserted statement is false for every $a \geq 7$, with its last obstruction at $n = a - 2$. Granting the source's own exclusion of the degree-2 cell, $a = 0$ and $2 \leq a \leq 6$ do satisfy the asserted range; we assert nothing at the degenerate index $a = 1$. What survives is proved rather than refuted: polynomiality, and degree exactly $\lfloor a/2 \rfloor$, hold for all $a \geq 2$, and the repaired hypothesis is $n \geq \max(\lfloor a/2 \rfloor + 2,\, a - 1)$.

1. The statement, with its locator

Let $M_{2,n}$ be the $2 \times n$ Miura-ori crease pattern and $\mathrm{OFG}(M_{2,n})$ its origami flip graph, whose vertices are the flat-foldable mountain–valley assignments of $M_{2,n}$ and whose edges join two assignments differing by a face flip. Following Christensen, Hull, O'Neil, Pappano, Ter-Saakov and Yang [1], v_n^d denotes the number of vertices of degree d in $\mathrm{OFG}(M_{2,n})$, and we abbreviate the *reverse diagonal*

$$f_a(n) \ := \ v_n^{2n-a} \qquad (a \in \mathbb{N}_0).$$

Our target is the theorem at lines 1256–1258 of the arXiv e-print LaTeX source `Miuradraft-v2.tex` of [1]; we call it the *reverse-diagonal theorem* below. We quote from the source file rather than the PDF so that the locator is exact.

```
\begin{theorem} \label{thm:reversepolynomials}
The explicit formula for $v_n^{2n-a}$ where $n \ge
\left\lceil \frac{a}{2} \right\rceil+1$ is a polynomial
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of $n$ with degree $\left\lfloor \frac{a}{2}
\right\rfloor$ for all $a \in \N_0$.
\end{theorem}
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(The source lines have been re-wrapped to fit this page; not a character of the statement is altered.) The sentence that introduces the theorem, at line 1254, reads:

> "The data in Table `\ref{tab:reversepolynomialdata}` suggests the following Theorem, which can be proven using Proposition `\ref{prop:recurrences}` and induction."

The environment then closes with no proof. Two further lines of the same file are used below and are quoted here because our argument rests on them.

- Line 1236 fixes the index set and states a carve-out in one paragraph: the reverse sequences are collected "for all $a \in \mathbb{N}_0$, where $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$", "this time excluding the edge case where there are exactly 4 vertices labeled with $2n-a$ (i.e. $2n-a=2$)".
- Line 1204 records the anomaly $v_1^2 = 2$, against $v_m^2 = 4$ for $m \geq 2$.
- The source's recurrence proposition of lines 1189–1202, which it *proves* there, gives

$$b_n^d = v_{n-1}^{d-2}, \qquad w_n^d = v_{n-1}^d + w_{n-1}^{d-1} + v_{n-2}^{d-2},$$
$$v_n^d = v_{n-1}^d + w_{n-1}^{d-1} + v_{n-1}^{d-2} + v_{n-2}^{d-2}.$$

Eliminating w from the last two identities (as [1] itself does at line 1219) puts the recurrence in pure-v form:

$$v_n^d \;=\; v_{n-1}^d + v_{n-1}^{d-1} + v_{n-1}^{d-2} + v_{n-2}^{d-2} - v_{n-2}^{d-3} \qquad (n \geq 3), \tag{1}$$

with the conventions of [1]: $v_m^e = 0$ for $e < 2$ or $e > 2m$, $v_1^2 = 2$, and $v_m^2 = 4$ for $m \geq 2$. We take (1) as given; it is a consequence of a proposition the source proves.

Two readings of the admissible start. Since $2n - a = 2$ has an integer solution only for even a, namely at $n = a/2 + 1 = \lceil a/2 \rceil + 1$, the theorem's literal first admissible n is itself the cell excluded at line 1236 when a is even. Read literally the theorem therefore already fails at $a = 6$, where the claimed degree is $\lfloor 6/2 \rfloor = 3$ and the first admissible index is $n = 4$: taking $f_6(4) = v_4^2 = 4$ from the convention at line 1204,

$$f_6(4), \ldots, f_6(8) \;=\; 4,\ 36,\ 128,\ 292,\ 544,$$

whose fourth forward difference is $544 - 4 \cdot 292 + 6 \cdot 128 - 4 \cdot 36 + 4 = 4 \neq 0$. That is not what we rest on. Throughout what follows we instead grant the source its own exclusion clause and work from

$$s(a) \;:=\; \left\lfloor \tfrac{a}{2} \right\rfloor + 2, \tag{2}$$

which for *odd a* equals the literal bound $\lceil a/2 \rceil + 1$ and for *even a* is one larger. That makes the refutation harder, and it makes the $a = 7$ witness below independent of any quantifier quibble. Consistently with (2), the printed rows of the source's reverse-polynomial data table begin at $n = \lfloor a/2 \rfloor + 2$ in all six of the rows $a = 0, \ldots, 5$ it displays.

2. The witness at $a = 7$

At $a = 7$ the claimed degree is $\lfloor 7/2 \rfloor = 3$, the claimed range is $n \geq \lceil 7/2 \rceil + 1 = 5 = s(7)$, and the exclusion clause is vacuous, since $2n - 7 = 2$ has no integer solution.

Theorem 1. *The reverse-diagonal theorem of* [1] *quoted above is false at* $a = 7$.

Proof. The five values

$$f_7(5), f_7(6), f_7(7), f_7(8), f_7(9) \;=\; 12,\ 88,\ 296,\ 680,\ 1288 \tag{3}$$

are cells of the source's own degree-distribution table. Their Newton forward-difference table is

f	12	88	296	680	1288
Δ^1		76	208	384	608
Δ^2			132	176	224
Δ^3				44	48
Δ^4					**4**

A function agreeing on five consecutive integers with a polynomial of degree at most 3 has vanishing fourth forward difference there. Here it is $4 \neq 0$, and the window $n = 5, \ldots, 9$ lies entirely inside the asserted range $n \geq 5$. $\square$

Remark. The five integers of (3) are not a printing accident: they are forced by (1) from other cells of the same table,

$$\begin{aligned}
v_5^3 &= 8 + 4 + 0 + 0 - 0 = 12,\\
v_6^5 &= 36 + 36 + 12 + 8 - 4 = 88,\\
v_7^7 &= 80 + 128 + 88 + 36 - 36 = 296,\\
v_8^9 &= 140 + 292 + 296 + 80 - 128 = 680,\\
v_9^{11} &= 216 + 544 + 680 + 140 - 292 = 1288.
\end{aligned}$$

The cells consumed here are recorded in Table 1.

3. The infinite family

The reverse-diagonal recurrence. Substituting $d = 2n - a$ into (1) and using $f_a(n) = v_n^{2n-a}$,

$$f_a(n) = f_a(n-1) + f_{a-1}(n-1) - f_{a-1}(n-2) + f_{a-2}(n-1) + f_{a-2}(n-2) \tag{4}$$

for every $n \geq 3$. Put $F_a(x) := \sum_{n \geq 1} f_a(n) x^n$. Multiplying (4) by x^n and summing over $n \geq 3$ leaves the boundary polynomial $f_a(1)x + [f_a(2) - f_a(1) -$

TABLE 1. Cells of the degree-distribution table of [1] quoted in this note. A dash is a printed blank, read as 0. Rows are indexed by the degree d and columns by $n = 2, \dots, 9$.

$d\backslash n$	2	3	4	5	6	7	8	9
3	0	4	8	12	16	20	24	28
5	–	0	8	36	88	168	280	428
7	–	–	0	12	80	296	792	1744
9	–	–	–	0	16	140	680	2396
11	–	–	–	–	0	20	216	1288

Additionally: the whole $n = 3$ column, $v_3^2 = 4$, $v_3^3 = 4$, $v_3^4 = 8$, $v_3^5 = 0$, $v_3^6 = 2$; the $a = 6$ reverse diagonal $v_5^4, v_6^6, v_7^8, v_8^{10}, v_9^{12} = 36, 128, 292, 544, 900$; and $v_4^2 = 4$ from the convention at line 1204.

$f_{a-1}(1) - f_{a-2}(1)]x^2$. Now $f_a(1) = v_1^{2-a} = 0$ for $a \geq 1$ and $f_a(2) = v_2^{4-a} = 0$ for $a \geq 3$, so *for every $a \geq 3$ the boundary vanishes identically* and

$$(1-x)F_a \;=\; x(1-x)F_{a-1} \;+\; x(1+x)F_{a-2} \qquad (a \geq 3), \tag{5}$$

with the two seeds, obtained from the $a = 2$ instance of (4) whose boundary term $2x^2$ is not zero,

$$F_0 = \frac{2x}{1-x}, \qquad F_1 = 0, \qquad F_2 = \frac{4x^2}{(1-x)^2}, \qquad F_3 = \frac{4x^3}{(1-x)^2}.$$

Equivalently $f_2(n) = 4(n-1)$ and $f_3(n) = 4(n-2)$ exactly, which reproduces the $a = 2$ and $a = 3$ rows of the source's reverse-polynomial data table.

The normalised numerator. Write $k_a := \lfloor a/2 \rfloor$ and

$$G_a(x) \;:=\; (1-x)^{k_a+1} F_a(x).$$

Since $k_a - k_{a-1} = 1$ and $k_a - k_{a-2} = 1$ for even a, while $k_a - k_{a-1} = 0$ and $k_a - k_{a-2} = 1$ for odd a, dividing (5) by $(1-x)^{k_a+1}$ leaves no remainder:

$$\begin{aligned} G_a &= x(1-x)G_{a-1} + x(1+x)G_{a-2} && (a \text{ even}),\\ G_a &= x\,G_{a-1} + x(1+x)G_{a-2} && (a \text{ odd}), \end{aligned} \tag{6}$$

with $G_2 = 4x^2$ and $G_3 = 4x^3$. Hence every G_a, $a \geq 2$, lies in $\mathbb{Z}[x]$; explicitly

$$G_4 = 4x^3 + 8x^4 - 4x^5, \qquad G_5 = 8x^4 + 12x^5 - 4x^6,$$
$$G_6 = 4x^4 + 20x^5 + 8x^6 - 20x^7 + 4x^8,$$
$$G_7 = 12x^5 + 40x^6 + 16x^7 - 24x^8 + 4x^9,$$

$$G_8 = 4x^5 + 36x^6 + 56x^7 - 36x^8 - 56x^9 + 32x^{10} - 4x^{11}.$$

The dictionary. For any $F = \sum_n f(n)x^n$ with $f(n) := 0$ for $n \leq 0$, and any $j \geq 0$,

$$[x^n](1-x)^j F(x) = (\Delta^j f)(n-j), \tag{7}$$

because $[x^n](1-x)^j F = \sum_i (-1)^i \binom{j}{i} f(n-i)$, and re-indexing $i = j-t$ turns this into $(-1)^j \sum_t (-1)^t \binom{j}{t} f(n-j+t) = (\Delta^j f)(n-j)$. With $j = k_a+1$ this reads

$$[x^n]G_a = (\Delta^{k_a+1} f_a)(n-k_a-1), \tag{8}$$

i.e. G_a *is* the (k_a+1)-st difference table of f_a, shifted.

The degree and the leading coefficient.

Proposition 2. *For $m \geq 1$, $\deg G_{2m} = 3m-1$ and $\deg G_{2m+1} = 3m$, and the leading coefficient of G_a is $L_a = 4(-1)^{\lfloor a/2\rfloor+1}$ for every $a \geq 2$. Moreover $G_a(1) > 0$ for every $a \geq 2$.*

Proof. Let $D_a = \deg G_a$. In (6) the competing leading terms are $-L_{a-1}x^{D_{a-1}+2}$ against $L_{a-2}x^{D_{a-2}+2}$ for even a, and $L_{a-1}x^{D_{a-1}+1}$ against $L_{a-2}x^{D_{a-2}+2}$ for odd a. From $D_2 = 2$, $D_3 = 3$ the claimed formula follows by induction, and in both parities the first candidate *strictly* dominates: for even a one has $D_{a-1} = D_{a-2}+1$, hence $D_{a-1}+2 > D_{a-2}+2$; for odd a one has $D_{a-1}+1 = D_{a-2}+3 > D_{a-2}+2$. So no cancellation can occur, and $L_a = -L_{a-1}$ for even a, $L_a = L_{a-1}$ for odd a. With $L_2 = L_3 = 4$ this gives $L_a = 4(-1)^{\lfloor a/2\rfloor+1}$; in particular $|L_a| = 4$ for all $a \geq 2$. Setting $x = 1$ in (6) kills the $x(1-x)$ term, so $G_a(1) = 2G_{a-2}(1)$ for even a and $G_a(1) = G_{a-1}(1) + 2G_{a-2}(1)$ for odd a; with $G_2(1) = G_3(1) = 4 > 0$, induction gives $G_a(1) > 0$. ☐

Corollary 3. *For every $a \geq 2$: f_a agrees with a polynomial of degree exactly $\lfloor a/2\rfloor$ on $[a-1,\infty)$, and $a-1$ is the least such threshold. The last nonvanishing value of $\Delta^{k_a+1}f_a$ is at base $n = a-2$, where it equals $4(-1)^{\lfloor a/2\rfloor+1}$.*

Proof. By (8), $\Delta^{k_a+1}f_a(n) = [x^{n+k_a+1}]G_a$, which vanishes exactly when $n+k_a+1 > D_a$, i.e. when $n > D_a-k_a-1$. Proposition 2 gives $D_a-k_a = a-1$ in both parities ($3m-1-m = 2m-1$ and $3m-m = 2m$), so $\Delta^{k_a+1}f_a$ vanishes on $[a-1,\infty)$ and equals L_a at $n = a-2$. Vanishing of the (k_a+1)-st difference on a half-line is agreement there with a polynomial of degree $\leq k_a$; the degree is exactly k_a because $\Delta^{k_a}f_a$ is eventually the constant $G_a(1) \neq 0$, again by (8) with $j = k_a$ and Proposition 2. The leading coefficient of that polynomial is $G_a(1)/k_a!$. ☐

Theorem 4. *The reverse-diagonal theorem of* [1] *is false for every $a \geq 7$. The base indices n inside the asserted range at which $\Delta^{\lfloor a/2\rfloor+1}f_a(n) \neq 0$ all lie in $\{s(a),\dots,a-2\}$, and the largest of them is $n = a-2$, which carries the value $4(-1)^{\lfloor a/2\rfloor+1}$. For $2 \leq a \leq 6$, by contrast, every nonvanishing value of $\Delta^{\lfloor a/2\rfloor+1}f_a$ lies strictly below $s(a)$, so on the range $n \geq s(a)$ granted by the source's exclusion clause those five indices satisfy the theorem. Taken* *literally, with first admissible index $\lceil a/2 \rceil + 1$, the theorem fails at $a = 6$ as well; see Section 1.*

Proof. By Corollary 3 the obstruction at $n = a - 2$ is nonzero for every $a \geq 2$, and it sits inside the range $n \geq s(a)$ precisely when

$$a - 2 \;\geq\; \left\lfloor \frac{a}{2} \right\rfloor + 2 \iff \left\lceil \frac{a}{2} \right\rceil \;\geq\; 4 \iff a \geq 7.$$

For those a the window $n = a-2, \ldots, a-2+k_a+1$ lies in the asserted range, and $\Delta^{k_a+1} f_a(a-2) \neq 0$ contradicts agreement there with a polynomial of degree $\leq \lfloor a/2 \rfloor$. Conversely for $2 \leq a \leq 6$ the same inequality shows $a - 2 < s(a)$, so every nonvanishing value of $\Delta^{k_a+1} f_a$ lies strictly below the first admissible n and the statement holds on the whole of $n \geq s(a)$. $\square$

Remark. This explains the source's own confirming window rather than merely contradicting it. The caption of its reverse-polynomial data table (line 1250) says: "The code computed all sequences up to $a = 16$, but only confirmed polynomial patterns up to $a = 6$." The obstruction exists for every $a \geq 2$, but $a - 2 < \lfloor a/2 \rfloor + 2$ exactly when $a \leq 6$, so for those a it lies below $s(a)$ and is invisible to any check confined to the range. The two cut-offs are the same inequality $\lceil a/2 \rceil \geq 4$.

4. What survives, and the repaired hypothesis

Proposition 2 and Corollary 3 prove the half of the reverse-diagonal theorem that the source only asserted:

Corollary 5. *For every $a \geq 2$, f_a agrees on a half-line with a polynomial of degree exactly $\lfloor a/2 \rfloor$ and leading coefficient $G_a(1)/\lfloor a/2 \rfloor !$. The same holds for $a = 0$ by inspection, since $f_0(n) = v_n^{2n} = 2$.*

Consequently, replacing the range clause of that theorem by

$$n \;\geq\; \max\left(\left\lfloor \frac{a}{2} \right\rfloor + 2,\; a - 1 \right) \tag{9}$$

makes it true for every $a \geq 2$, and for $a = 0$; the degenerate index $a = 1$ is outside anything claimed here. The repair also supersedes the even-a boundary question, since $\lfloor a/2 \rfloor + 2$ excludes the degree-2 cell in both parities. At $a = 7$ the repaired bound is 6; the two terms record the admissible start (2) and the agreement threshold of Corollary 3 respectively. In particular the weaker repair $n \geq \lfloor a/2 \rfloor + 2$ is *not* correct: it is 5 at $a = 7$, which is the very threshold Theorem 1 refutes.

Status. What is refuted is the *range clause* of the reverse-diagonal theorem, as a statement quantified over all $a \in \mathbb{N}_0$ and all $n \geq \lceil a/2 \rceil + 1$. Polynomiality and the degree $\lfloor a/2 \rfloor$ are *not* refuted: they are true for every $a \geq 2$ and are proved above, where the source asserts them without proof. Nothing here touches the source's proved sibling theorem at lines 1208–1226 (fixed degree d), which our program corroborates at $m = 2$ for $d = 2, \ldots, 40$ on a sampled window $n = \lceil d/2 \rceil + 1, \ldots, 85$ of its stated range, finding no obstruction there. The defect is specific to the reverse diagonal, whose range clause is the sibling's word for word with d replaced by a.

What is not settled.

- The source's recurrence proposition is *not* re-proved here. It is proved in [1] at lines 1191–1202 and taken as given; what we add is the w-elimination (1) and the boundary conventions, which are pinned against the source's own printed integers (Table 1) rather than re-derived from the degree-extension-forest construction. If that proposition failed at a boundary not exercised by those cells, (4) would need redoing.
- No stage of this work brute-forced mountain–valley assignments of $M_{2,n}$; the flat-foldability model is trusted, legitimately, because the source proves the recurrence it comes from. Theorem 1 does not depend on the model at all, since its five values are printed cells. The family $a \geq 8$, and every value at $n \geq 10$, does.
- Nothing is claimed about the theorem at the degenerate index $a = 1$. The seed $F_1 = 0$ above records that the reverse diagonal there is the zero sequence (the census reproduces $f_1(n) = 0$ for $n \leq 39$), so whether the asserted "degree exactly $\lfloor 1/2 \rfloor = 0$" is to count as failing at $a = 1$ turns on the convention assigning a degree to the zero polynomial and on nothing else. We draw no conclusion from it, and the refutation does not use it.
- Beyond $a = 80$ and $n = 130$ nothing is checked by computation. Proposition 2, Corollary 3 and Theorem 4 are proved for all a; the accompanying census only corroborates them inside those bounds. Which indices of $\{s(a), \ldots, a-2\}$ other than $n = a-2$ actually fail is not asserted here.

Related work, and a prior threshold of the same shape. Gupta [2] enumerates degree-4 and degree-5 vertices of $\mathrm{OFG}(M_{m,n})$ by height functions, together with diameters; it says nothing about a reverse diagonal (its "antidiagonal" is the grid antidiagonal $\{i + j = s\}$ of a height function). Gupta [3] reached, in the *fixed-d bivariate* slicing of the same family, a validity region $m, n \geq \max(d-1, 2)$ with the count departing from the polynomial below it, and an onset magnitude 4: "one step below, this departure has leading coefficient -4 times a Baxter number" through $d = 11$. So neither the threshold shape "parameter minus one" nor the magnitude 4 is presented here as unprecedented. Neither work settles the reverse-diagonal slicing $d = 2n - a$: [3] contains no such slicing, at $m = 2$ its region $m, n \geq \max(d-1, 2)$ forces $d \leq 3$, and it treats the $m = 2$ results of [1] as correct and settled.

5. Verification

Theorem 1 needs no computer: five integers printed in [1] and ten subtractions. Theorem 4, Proposition 2, Corollary 3 and the repair (9) are hand algebra, printed above in full with $G_2, \ldots, G_8$ written out.

The accompanying program `verify.py` (Python 3.9+, standard library only, exact integer arithmetic throughout) re-derives the *finite* assertions of this note from the objects printed above: the seeds $v_1^2 = 2$, $v_2^2 = 4$, $v_2^3 = 0$, $v_2^4 = 2$; the recurrence (1); the cells of Table 1; the witness (3); the polynomials $G_2, \ldots, G_8$. It calibrates the recurrence against the source's printed integers before using it, verifies the dictionary (8) against the recurrence, checks the degree, leading-coefficient and positivity statements of Proposition 2, the threshold $a-1$ and the repair (9) for $a \leq 80$ — both from G_a and directly from difference tables — and censuses the source's proved sibling theorem as a null. It does *not* verify the universal arguments: the proofs of Proposition 2, Corollary 3 and Theorem 4 are hand proofs and are only corroborated there; the source's recurrence proposition is not re-proved; no mountain–valley assignment is enumerated anywhere; and the line numbers, byte count and md5 of the source file are not fetched. Its transcript, with its own closing statement of scope, is `verify.output.txt`.

References

[1] L. Christensen, T. C. Hull, E. O'Neil, V. Pappano, N. Ter-Saakov, and K. Yang, *The origami flip graph of the* $2 \times n$ *Miura-ori*, arXiv:2506.19700v2, 2026. Quotations and line numbers refer to the LaTeX source `Miuradraft-v2.tex` of the v2 e-print (148 285 bytes, md5 `adca23f5c2eade3979cfc77682341c6d`); we located no later e-print version and no journal reference for it.

[2] C. Gupta, *Height functions on the* $m \times n$ *Miura-ori flip graph: degree sequence and diameter*, arXiv:2606.22614v2, 2026.

[3] C. Gupta, *One construction for the Miura-ori flip-graph degree sequence*, arXiv:2607.05567v3, 26 August 2026.